

Age and Emotion Detection through Voice

1) Project Overview

This project implements an Age and Emotion Detection through Voice system, built using the same modelling approach as the Car Colour Detection and Traffic Analysis training project.

The system combines Deep Learning and Audio Signal Processing techniques to:

1. Predict a speaker's gender from a voice recording.
2. Reject the input with the message "Upload male voice." if the voice is not male.
3. Predict the speaker's age from the voice recording.
4. Apply an age-gated rule: for predicted ages above 60, mark the speaker as a senior citizen and additionally detect their emotion.
5. Provide a graphical user interface (GUI) for uploading or recording a voice note and viewing the result.

2) Problem Statement

Manually screening voice notes for a caller's age, gender, or emotional state doesn't scale. This task adds a deliberate age-dependent rule on top of a baseline gender and age prediction system.

The objective of this project is to develop a machine learning system capable of:

1. Predicting gender from a voice note using a CNN.
2. Rejecting non-male voices with an appropriate message.
3. Predicting age from a voice note using a CNN.
4. Detecting emotion only for speakers whose predicted age is above 60.
5. Displaying results through a user-friendly GUI.

3) Dataset

CREMA-D Dataset

Dataset Source:

Kaggle - [ejlok1/cremad](#)

Real ages for every actor are fetched live from the dataset's official demographics file, so no separate age-annotation step was required:

VideoDemographics.csv → columns ActorID, Age, Sex, Race, Ethnicity

Filenames follow the format: ActorID_Sentence_Emotion_Intensity.wav

Dataset Structure:

- 7,442 clips used, spanning 91 actors
- Real documented ages ranging from 20 to 74 years

- Only 6 of the 91 actors are over 60 years old
- After filtering to male-only voices for age training, only 5 senior actors remain out of 48 male actors

RAVDESS Dataset (supplementary)

Dataset Source:

Kaggle - uwrkaggler/ravdess-emotional-speech-audio

2,496 additional clips used to add more gender and emotion training examples. All RAVDESS actors are young/mid-life (roughly ages 21-33), so this dataset is deliberately excluded from age training and only strengthens gender/emotion training.

Total spectrograms cached for training across both datasets: 9,938 images.

4) Methodology

1. Data Preprocessing

- Audio loaded and resampled to 16kHz mono using Librosa.
- Silence trimmed; clips shorter than 1 second padded.
- Each clip converted into a mel-spectrogram, then to decibels, then normalized to a 0-255 image, exactly like a regular image.
- Spectrogram resized to 128 x 128 pixels using OpenCV, the same resize call used on cropped car images in the training project.
- Dataset split by actor (GroupShuffleSplit), not by clip, so reported accuracy reflects performance on voices the model never heard during training.

2. Gender Classification Model

A Convolutional Neural Network (CNN) was trained to classify gender, built the same way as the vehicle colour CNN in the training project: a stack of Conv2D + MaxPooling2D blocks, then Dense layers.

Model Architecture:

- Convolution Layer (32 filters)
- Max Pooling Layer
- Convolution Layer (64 filters)
- Max Pooling Layer
- Flatten Layer
- Dense Layer (64 units)
- Dense Output Layer (1 unit, Sigmoid)

3. Age Estimation Model

Reuses the exact same CNN block structure as the gender model above. Only the final layer changes, from a Sigmoid (yes/no) output to a Linear (numeric) output, since age is a continuous value rather than a category. Trained only on male voices with known real ages.

4. Emotion Detection Model

A deeper CNN (three Conv2D + MaxPooling2D blocks of 32, 64 and 128 filters, followed by a Dense layer) was trained to classify emotion into six classes: angry, disgust, fear, happy, neutral and sad. Trained on all available clips from both CREMA-D and RAVDESS.

5. Senior-Citizen Gated Rule

For each processed voice note:

1. Predict gender using the gender CNN.
2. If gender is not male, reject immediately with "Upload male voice."
3. If gender is male, predict age using the age CNN.
4. If the predicted age is above 60:
 - o Mark the speaker as a Senior Citizen
 - o Additionally predict emotion using the emotion CNN
- Else:
 - o Report the estimated age only, with no emotion prediction

5) GUI Development

Gradio was used to develop a graphical user interface.

Features:

- Upload or record a voice note directly in the browser
- Preview and play back the uploaded/recorded audio
- Predict gender using the gender CNN
- Reject the input if the voice is not male
- Predict age using the age CNN
- Apply the senior-citizen gate
- Detect emotion using the emotion CNN, only when the speaker is a senior citizen
- Display gender, estimated age, senior citizen status and (if applicable) emotion

6) Results

Gender Classification

Validation Accuracy: ~90%

Age Estimation

Test MAE: 11.6 years

Test R-squared: -0.48 (negative, due to the very small number of senior training examples)

Senior Citizen (60+) Threshold Accuracy: 91.5%

Emotion Detection

Validation Accuracy: ~46-50%

Pipeline Verification

Successfully verified on real dataset clips, confirming the full chain behaves as specified:

- A real female CREMA-D clip correctly returned "Upload male voice."
- Real male clips under 60 correctly returned gender and estimated age only, with no emotion prediction.
- The Gradio interface launched successfully with a public share link and produced correct results for both uploaded and recorded audio.

7) Visualizations

Included in notebook:

- Gender Model Training Accuracy Graph
- Gender Model Training Loss Graph
- Gender Confusion Matrix
- Age Model MAE Graph
- Age Model Training Loss Graph
- Emotion Model Training Accuracy Graph
- Emotion Model Training Loss Graph
- Emotion Confusion Matrix
- GUI Outputs

8) Technologies Used

- Python
- TensorFlow / Keras
- Librosa
- OpenCV
- NumPy
- Pandas
- Scikit-Learn
- Matplotlib
- Gradio

9) Future Improvements

- Collect more senior-voice training data to improve age estimation accuracy.
- Use early stopping or fewer epochs on the emotion model to reduce overfitting.
- Train on natural, non-acted speech for more realistic emotion detection.
- Add confidence scores to predictions.
- Support real-time voice stream processing instead of single clips.
- Deploy the application using Hugging Face Spaces, Streamlit, or Flask.

10) Conclusion

This project successfully extends the Car Colour Detection and Traffic Analysis training project with a new audio-based Deep Learning feature: an age-gated emotion detection system for voice notes. The developed system predicts gender and age from a voice recording, rejects non-male voices as specified, and detects emotion only for senior citizens above 60 years of age, displaying all results through an interactive Gradio GUI. The pipeline was verified against real dataset clips, confirming the gender-rejection rule and the senior-citizen emotion gate both work exactly as specified, while the age regression's known limitations (due to limited senior training data) are documented transparently.